

Liji Mariyam Mathew

AI/ML Engineer

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SUMMARY

AI/ML Engineer with 2+ years of production experience building multi-tenant SaaS platforms, LLM-integrated dashboards, and Airflow-orchestrated automation agents for regulated, security-sensitive domains. Primary engineer behind four production agents on lowtouch.ai's 44-agent **Agentomatic** platform — **ImageGuard** (banking-grade container compliance), **PV Signal Agent** (GxP pharmacovigilance case management), **FluxIQ/Controllershhip** (multi-company financial reconciliation with an embedded AI assistant), and an internal timesheet analytics agent — spanning React/Next.js frontends, FastAPI backends, Keycloak-based multi-tenant auth, and Apache Airflow pipelines. AWS Certified Machine Learning Associate.

SKILLS

Programming	Python, TypeScript, SQL
Gen AI & LLMs	LangChain, LangGraph, ReAct Agents, RAG, OpenAI GPT-4/GPT-5, Anthropic Claude, Google Gemini, Ollama, Grok/XAI, Prompt Engineering, Multi-Agent Orchestration, Dynamic Tool Generation from OpenAPI
AI/ML Frameworks	PyTorch, TensorFlow, Keras, Scikit-learn, LSTM, Random Forest, SARIMA
Data Engineering & Pipelines	Apache Airflow 3.x, ETL/ELT Pipeline Design, Data Pipeline Orchestration, PostgreSQL, pgvector, Chroma, MariaDB, MongoDB, OpenSearch, Redis
Backend & APIs	FastAPI, Flask, REST APIs, Pydantic, OpenAPI/Swagger, Google Sheets API
Frontend	Next.js 15 (App Router, React Server Components), React 19, TypeScript, Tailwind CSS, Chart.js, NextAuth, Vitest, WCAG accessibility
Security & Compliance	Trivy (CVE/license scanning), Keycloak OIDC, OAuth2, JWT, RBAC, HashiCorp Vault, GxP / 21 CFR Part 11
Observability	Prometheus, Grafana, Alertmanager, ELK Stack
Cloud & DevOps	AWS (EC2, S3, ECR, SageMaker), Docker, CI/CD (GitHub Actions), MinIO, Supervisor

EXPERIENCE

AI Developer — Lowtouch.ai

Nov 2024 – Present

- Largest single contributor to ImageGuard (148 of 285 commits), a multi-tenant container vulnerability and compliance platform for a banking client — built workspace-scoped multi-tenancy end to end (client-scoped headers, per-client data isolation, Keycloak auth), migrated the dashboard to React Server Components to eliminate the post-login fetch waterfall, and integrated an LLM-driven compliance chat assistant into the UI; authored 9 production Airflow DAGs (Trivy CVE/license scanning, AWS ECR image sync, compliance reporting) governing 17 product and 30 shared infrastructure images.
- Built the PV Signal Agent pharmacovigilance platform end to end — a Next.js/FastAPI case-management UI on an 11-DAG Apache Airflow pipeline (21 CFR Part 11, ICH E2B(R3)) that moves adverse-event reports from email intake through MedDRA coding to CDSCO ICSR submission; implemented the 3-stage human-in-the-loop workflow (Analyst → Medical Reviewer → Supervisor) with SLA countdown tracking and live, Airflow-XCom-driven pipeline progress visualization.

- Built FluxIQ (Controllershship Agent) from its initial scaffold — a multi-company financial reconciliation dashboard with Keycloak-backed RBAC (custom realm role mapping), an embedded AI assistant for balance/flux/completeness queries, batch generate/feed workflows with live progress tracking and DAG-failure attribution, a full audit trail with per-report version history, and a Vitest test suite.
- Built an internal Timesheet & Performance Tracker agent automating weekly/monthly reporting — PDF classification, performance-band scoring, SMTP-based report distribution, and ranking APIs.
- Core contributor to Agentomatic, lowtouch.ai's 44-agent LLM orchestration platform — own 4 of its production agents (above) end to end using Claude Code CLI for AI-assisted full-stack delivery; also contributed the initial Next.js scaffold for InvoFlux (AP automation) and Gemini/Ollama integration work on the ClipFoundry media pipeline.

Data Scientist — Gofreelab Technologies Pvt. Ltd.

May 2024 – Nov 2024

- Built supervised ML models (Random Forest, Logistic Regression, 80%+ accuracy) and an NLP sentiment classifier (93.93% accuracy); served via Flask/FastAPI on AWS EC2 for real-time inference.
- Automated end-to-end ML pipelines — web scraping, API ingestion, preprocessing, training, deployment — including a hybrid SARIMA + Random Forest stock forecasting model (97%+ accuracy).

KEY PROJECTS

ImageGuard · Lowtouch.ai — Multi-tenant container vulnerability & compliance platform for a banking client; largest contributor (148/285 commits). Next.js 15 (RSC), FastAPI, Airflow 3.x, Trivy, PostgreSQL, AWS ECR, Keycloak, Vault, Prometheus/Grafana.

PV Signal Agent · Lowtouch.ai — GxP pharmacovigilance case-management platform: 11-DAG pipeline, 3-stage HITL review, live Airflow-XCom progress tracking. Next.js 15, FastAPI, Airflow 3.x, LangChain/LangGraph, pgvector, Keycloak OIDC.

FluxIQ (Controllershship Agent) · Lowtouch.ai — Multi-company financial reconciliation dashboard built from initial scaffold; embedded AI assistant, RBAC, audit/version history. Next.js 15, Keycloak RBAC, Airflow 3.x, Vitest, Docker.

Tracker Agent · Lowtouch.ai — Automated timesheet/performance reporting: PDF classification, band-scoring, SMTP distribution, ranking APIs. Python, Apache Airflow, SMTP.

Agentomatic · Lowtouch.ai — 44-agent LLM orchestration platform with dynamic OpenAPI tool generation; personally own 4 production agents (above). Python, FastAPI, LangChain/LangGraph, Chroma/pgvector, Vault.

EDUCATION

M.Sc. Applied Statistics with Data Analytics

Mahatma Gandhi University, Kerala
Aug 2022 – Mar 2024

B.Sc. Mathematics

Mahatma Gandhi University, Kerala
Jun 2019 – Apr 2022

CERTIFICATIONS

AWS Certified Machine Learning - Associate (MLA-C01)

Amazon Web Services
Issued Jan 2026 · Expires Jan 2029